

姓名：

(號)

成績

年級：

科目：

練習題 16.1

1. 因為 f 在 $[0, 1]$ 內可積, 所以 $\exists m_f, M_f \in \mathbb{R}$, 使得 $\forall x \in [0, 1]$, $m_f \leq f(x) \leq M_f$
 同理可得 $\exists m_g, M_g \in \mathbb{R}$, 使得 $\forall y \in [0, 1]$, 都有 $m_g \leq g(y) \leq M_g$

所以記 $F(x) = f(x) - m_f$, $G(y) = g(y) - m_g$

顯然 F, G 在 $[0, 1]$ 內皆可積, 記 $\int_0^1 F(x) dx = A$, $\int_0^1 G(y) dy = B$

因此 $\forall \varepsilon > 0$, $\exists \delta > 0$, 對於任意分割 π_x , 只要 $\|\pi_x\| < \delta$, 都有 $\bar{S}_F(\pi_x) - \underline{S}_F(\pi_x) < \varepsilon$
 對於任意分割 π_y , 只要 $\|\pi_y\| < \delta$, 都有 $\bar{S}_G(\pi_y) - \underline{S}_G(\pi_y) < \varepsilon$

記 $\pi_x: 0 = x_0 < x_1 < \dots < x_n = 1$

$\pi_y: 0 = y_0 < y_1 < \dots < y_m = 1$

$$\begin{aligned} & \sum_{i=1}^n \sum_{j=1}^m F(\xi_i) G(\eta_j) \Delta x_i \Delta y_j \\ &= \sum_{i=1}^n \left(F(\xi_i) \Delta x_i \left(\sum_{j=1}^m G(\eta_j) \Delta y_j \right) \right) = \left(\sum_{i=1}^n F(\xi_i) \Delta x_i \right) \left(\sum_{j=1}^m G(\eta_j) \Delta y_j \right) \end{aligned}$$

$$\text{所以 } \underline{S}_F(\pi_x) \underline{S}_G(\pi_y) \leq \sum_{i=1}^n \sum_{j=1}^m F(\xi_i) G(\eta_j) \Delta x_i \Delta y_j \leq \bar{S}_F(\pi_x) \bar{S}_G(\pi_y)$$

$$\text{令 } \|\pi_x\|, \|\pi_y\| \rightarrow 0, \text{ 則有 } \iint_I F(x) G(y) dx dy = \left(\int_0^1 F(x) dx \right) \left(\int_0^1 G(y) dy \right)$$

$$\Leftrightarrow \iint_I f(x) g(y) dx dy = \iint_I m_f g(y) dx dy + \iint_I f(x) m_g dx dy + \iint_I m_f m_g dx dy$$

$$= \left(\int_0^1 f(x) dx \right) \left(\int_0^1 g(y) dy \right) - m_f \left(\int_0^1 g(y) dy \right) - m_g \left(\int_0^1 f(x) dx \right) + m_f m_g$$

$$\Leftrightarrow \iint_I f(x) g(y) dx dy = \left(\int_0^1 f(x) dx \right) \left(\int_0^1 g(y) dy \right)$$

$$2. \iint_{[0,1] \times [0,1]} e^{x+y} dx dy = \iint_{[0,1] \times [0,1]} e^x \cdot e^y dx dy = \left(\int_0^1 e^x dx \right) \left(\int_0^1 e^y dy \right) = (e-1)^2$$

$$3. \text{ 記 } \pi_x: -a = x_0 < x_1 < \dots < x_n = a$$

$$\pi_y: -a = y_0 < y_1 < \dots < y_m = a$$

$$\begin{aligned} \sum_{i=1}^n \sum_{j=1}^m \sin(\xi_i + \eta_j) \Delta x_i \Delta y_j &= \left(\sum_{i=1}^n \sum_{j=1}^m \sin(\xi_i) \cos(\eta_j) \Delta x_i \Delta y_j \right) + \left(\sum_{i=1}^n \sum_{j=1}^m \sin(\eta_j) \cos(\xi_i) \Delta x_i \Delta y_j \right) \\ &= \left(\sum_{i=1}^n \sin(\xi_i) \Delta x_i \right) \left(\sum_{j=1}^m \cos(\eta_j) \Delta y_j \right) + \left(\sum_{i=1}^n \cos(\xi_i) \Delta x_i \right) \left(\sum_{j=1}^m \sin(\eta_j) \Delta y_j \right) \end{aligned}$$

令 $\|\pi_x\| \rightarrow 0, \|\pi_y\| \rightarrow 0$, 則可得

$$\begin{aligned} \iint_{[-a,a] \times [-a,a]} \sin(x+y) dx dy &= \left(\int_{-a}^a \sin x dx \right) \left(\int_{-a}^a \cos x dx \right) + \left(\int_{-a}^a \cos x dx \right) \left(\int_{-a}^a \sin y dy \right) \\ &= 0 \end{aligned}$$

幾何上來看, 若 $(x, y) \in I$, 則 $(-x, -y) \in I$

$$\sin(x+y) + \sin(-x-y) = 0$$

所以這個圓柱體關於 xy 平面對稱。

$$\begin{aligned} 4. \sum_{i=1}^k (M_i - m_i) \sigma(I_i) &= \sum_{i=1}^k M_i \sigma(I_i) - \sum_{i=1}^k m_i \sigma(I_i) \\ &= \bar{S}(\pi) - \underline{S}(\pi) < \varepsilon \end{aligned}$$

$\Leftrightarrow f$ 在 I 上可積

5. 考慮分割 $\pi: I, I_1 \cup \dots \cup I_k$

$$\text{記 } \sup_{x \in I_i} f(x) = M_i, \inf_{x \in I_i} f(x) = m_i, \sup_{x \in I_i} |f(x)| = M_i', \inf_{x \in I_i} |f(x)| = m_i'$$

會出現 $m_i \leq m_i' \leq M_i \leq M_i'$ 或 $m_i \leq M_i \leq m_i' \leq M_i'$

(i) 若 $m_i \geq 0$ 或 $M_i < 0$, 則 $M_i - m_i = M_i' - m_i'$

(ii) 若 $M_i \geq 0$ 且 $m_i < 0$, 則

(I.) 如果 $M_i = M_i'$, 則 $M_i' - m_i' = M_i - m_i' \leq M_i - m_i$

(II.) 如果 $M_i < M_i'$, 則 $M_i' = -m_i$, $M_i' - m_i' = -(m_i + m_i') \leq -m_i \leq M_i - m_i$

$$\text{所以有 } \sum_{i=1}^k (M_i' - m_i') \sigma(I_i) \leq \sum_{i=1}^k (M_i - m_i) \sigma(I_i)$$

姓名：

(號)

成績

年級：

科目：

因為 f 在 I 上可積，所以 $\forall \varepsilon > 0, \exists \delta > 0$ ，只要分割 π 滿足 $\|\pi\| < \delta$ 就有 $\sum_{k=1}^n (M_k - m_k) \delta(I_k) < \varepsilon$

因此， $|f|$ 在 I 上也可積

另一方面，
$$\left| \sum_{k=1}^n f(\xi_k) \delta(I_k) \right| \leq \sum_{k=1}^n |f(\xi_k)| \delta(I_k)$$

令 $\|\pi\| \rightarrow 0$ ，可得

$$\left| \int_I f d\sigma \right| \leq \int_I |f| d\sigma$$

練習題 16.2.

1. 設 $\vec{p} = \lim_{n \rightarrow \infty} \vec{p}_n$ ，那麼 $\forall \varepsilon \in (0, 1)$ ， $\exists N \in \mathbb{N}$ ， $\forall n > N$ ，都有 $\|\vec{p}_n - \vec{p}\| < \frac{\varepsilon}{2}$

記 $I = \left(\bigcup_{k=1}^N B_{\frac{\varepsilon}{2^{k+1}}}(\vec{p}_k) \right) \cup B_{\frac{\varepsilon}{2}}(\vec{p})$ 。因為 $\{\vec{p}_n : n \in \mathbb{N}\} \subset I$ ，而且 $\sigma(I) = \sum_{k=1}^N \pi \left(\frac{\varepsilon}{2^{k+1}} \right)^2 + \pi \left(\frac{\varepsilon}{2} \right)^2$

$$< \pi \sum_{k=1}^{N+1} \frac{\varepsilon^2}{2^k} < \pi \varepsilon$$

所以 $\{\vec{p}_n\}$ 是零面積集

3. $[0, 1]^2$ 的全體有理點集是 A 。 $\forall (x, y) \in [0, 1]^2$ ， $\exists \{x_n\}, \{y_n\} \subset [0, 1] \cap \mathbb{Q}$ ， $\lim_{n \rightarrow \infty} x_n = x$ ， $\lim_{n \rightarrow \infty} y_n = y$
 $\Rightarrow$ 對於 $\{(x_n, y_n)\} \subset A$ ， $\lim_{n \rightarrow \infty} (x_n, y_n) = (x, y) \Leftrightarrow (x, y) \in A'$

$$\Rightarrow \bar{A} = A \cup A' = A \cup [0, 1]^2 = [0, 1]^2, \quad \sigma(\bar{A}) = 1^2 = 1 \Rightarrow \bar{A} \text{ 不是零面積集} \Leftrightarrow A \text{ 不是零面積集}$$

因為 $[0, 1] \cap \mathbb{Q}$ 是可數集，所以可記 $[0, 1] \cap \mathbb{Q} = \{b_1, b_2, \dots, b_n, \dots\} = B$

$$A = B^2 = \{b_n\}_{n=1}^\infty \times \{b_n\}_{n=1}^\infty$$

$$\forall (b_i, b_j) \in A, \text{ 取開集 } I_{i,j}, \text{ 使得 } (b_i, b_j) \in I_{i,j} \text{ 且 } \sigma(I_{i,j}) = \frac{\varepsilon}{(i+j-1)2^{i+j-1}} \quad (\forall \varepsilon > 0)$$

$$\text{所以 } A \subset \bigcup_{i,j \in \mathbb{N}^+} I_{i,j}, \text{ 而且 } \sigma\left(\bigcup_{i,j \in \mathbb{N}^+} I_{i,j}\right) = \sum_{i \in \mathbb{N}^+} \sum_{j \in \mathbb{N}^+} \frac{\varepsilon}{(i+j-1)2^{i+j-1}} = \sum_{n \in \mathbb{N}^+} \frac{\varepsilon}{2^n} = \varepsilon$$

$\Rightarrow A$ 是零測集

4. 記 $D_J(f)$ 是 f 在 J 上的不連續點集, $D_I(f)$ 是 f 在 I 上的不連續點集
 因為 f 在 I 上可積, 所以 $D_I(f)$ 是零測集且 f 在 I 上有界
 因為 $J \subset I$, 所以 $D_J(f) \subset D_I(f)$ 且 f 在 J 上有界 $\Leftrightarrow f$ 在 J 上可積

5. 若 $\int_I f d\sigma = 0$, 則 $\int_I f d\sigma = 0$. 對任一分割 $\pi: I_1 \cup I_2 \cup \dots \cup I_k$

$$0 < \underline{S}(\pi) \leq \int_I f d\sigma = 0 \text{ 矛盾} \Rightarrow \int_I f d\sigma > 0$$

$$8. D(fg) = \{(x, y): (x, y) \in D(f) \cap D(g)\} \cup \{(x, y): (x, y) \in D(f) \setminus D(g)\} \cup \{(x, y): (x, y) \in D(g) \setminus D(f)\} \\ = D(f) \cup D(g)$$

因為 f, g 在 I 上可積, 所以 $D(f), D(g)$ 都是零測集且 f, g 在 I 上有界
 $\Rightarrow D(fg) = D(f) \cup D(g)$ 也是零測集且 fg 在 I 上有界
 $\Leftrightarrow fg$ 在 I 上可積

另一方面, $\forall x \in I, g(x) \neq 0 \Rightarrow D(1/g) = D(g) \Leftrightarrow g$ 在 I 上可積
 所以 $1/g$ 在 I 上可積

問題 16.2.

2. 若 $\forall J \subset I$, 都有 $f \leq 0$, 則 $\int_I f d\sigma \leq 0$
 所以若 $\int_I f d\sigma > 0$, 則 $\exists J \subset I$, 使得 $f > 0$

練習題 16.3

$$1. (1) \int_0^1 \int_0^1 \frac{x^2}{1+y^2} dx dy = \int_0^1 dy \int_0^1 \frac{x^2}{1+y^2} dx = \left(\int_0^1 \frac{1}{1+y^2} dy \right) \left(\int_0^1 x^2 dx \right) = \left(\arctan y \Big|_0^1 \right) \left(\frac{1}{3} x^3 \Big|_0^1 \right) = \frac{\pi}{12}$$

$$(2) \int_0^{\pi} \int_0^1 x \cos(xy) dx dy = \int_0^{\pi} dx \int_0^1 x \cos(xy) dy = \int_0^{\pi} \sin x dx = -\cos x \Big|_0^{\pi} = 1$$

$$(3) \int_0^{\pi} \int_0^{\pi} \sin(x+y) dx dy = \left(\int_0^{\pi} \sin x dx \right) \left(\int_0^{\pi} \cos y dy \right) + \left(\int_0^{\pi} \cos x dx \right) \left(\int_0^{\pi} \sin y dy \right) = 0$$

姓名：

(號)

地 址

成績

年級：

科目：

$$2. \iint_D \frac{\partial^2 f}{\partial x \partial y}(x, y) dx dy = \int_c^d dy \int_a^b \frac{\partial^2 f}{\partial x \partial y}(x, y) dx = \int_c^d \frac{\partial f}{\partial y}(b, y) dy - \int_c^d \frac{\partial f}{\partial y}(a, y) dy$$

$$= f(b, d) - f(b, c) - f(a, d) + f(a, c)$$

$$3. (1) \int_1^9 f d\sigma = \iint_D f(x, y) dx dy = \int_0^1 dx \int_0^1 f(x, y) dy = \int_0^1 dx \int_0^1 1 dy = \int_0^1 x^2 dx = \frac{1}{3}$$

$$(2) \int_1^9 f d\sigma = \iint_D f(x, y) dx dy = \int_0^1 dx \int_{\frac{\sqrt{2}}{2}}^1 f(x, y) dy = \int_0^{\frac{\sqrt{2}}{2}} dx \int_{x^2}^{2x^2} (x, y) dy + \int_{\frac{\sqrt{2}}{2}}^1 dx \int_{x^2}^1 (x, y) dy$$

$$= \int_0^{\frac{\sqrt{2}}{2}} \left(\frac{1}{2} (x+y)^2 \right) \Big|_{x^2}^{2x^2} dx + \int_{\frac{\sqrt{2}}{2}}^1 \left(\frac{1}{2} (x+y)^2 \right) \Big|_{x^2}^1 dx$$

$$= \frac{1}{2} \left[\int_0^{\frac{\sqrt{2}}{2}} [(2x^2+x)^2 - (x^2+x)^2] dx + \int_{\frac{\sqrt{2}}{2}}^1 [(x+1)^2 - (x^2+x)^2] dx \right]$$

$$= \frac{1}{2} \left[\int_0^{\frac{\sqrt{2}}{2}} x^2 (3x^2+2x) dx + \int_{\frac{\sqrt{2}}{2}}^1 (1-x^2)(x+1)^2 dx \right]$$

$$= \frac{1}{2} \left[\int_0^{\frac{\sqrt{2}}{2}} (3x^4+2x^3) dx + \int_{\frac{\sqrt{2}}{2}}^1 x(x+1)^3 dx - \int_{\frac{\sqrt{2}}{2}}^1 (x+1)^4 dx \right]$$

$$= \frac{1}{2} \left(\left[\frac{3}{5} x^5 + \frac{1}{2} x^4 \right]_0^{\frac{\sqrt{2}}{2}} + \frac{1}{2} (x+1)^4 \Big|_{\frac{\sqrt{2}}{2}}^1 - \frac{1}{5} (x+1)^5 \Big|_{\frac{\sqrt{2}}{2}}^1 \right)$$

$$= \frac{1}{2} \left(\frac{3\sqrt{2}}{40} + \frac{1}{8} + 8 - \frac{1}{2} \times \left(\frac{\sqrt{2}}{2} + 1 \right)^4 - \frac{32}{5} + \frac{1}{5} \left(\frac{\sqrt{2}}{2} + 1 \right)^5 \right)$$

$$= \frac{1}{2} \left(\frac{3\sqrt{2}+69}{40} - \left(\frac{1}{2} - \frac{\sqrt{2}}{10} - \frac{1}{5} \right) \left(1 + \frac{\sqrt{2}}{2} \right)^4 \right)$$

$$= \frac{1}{2} \left(\frac{3\sqrt{2}+69}{40} - \frac{3-\sqrt{2}}{10} \left(1 + \frac{\sqrt{2}}{2} \right)^4 \right)$$

練習題 16.4

$$2. \iint_{|x|+|y| \leq 10} \frac{dx dy}{100 + \cos^2 x + \cos^2 y} = \frac{1}{100 + \cos^2 \xi_x + \cos^2 \xi_y} \sigma(\{|x|+|y| \leq 10\}), \quad \sigma(\{|x|+|y| \leq 10\}) = (10+10) \times 10 = 200$$

$$\text{所以 } \frac{200}{102} \leq \iint_{|x|+|y| \leq 10} \frac{dx dy}{100 + \cos^2 x + \cos^2 y} < \frac{200}{100} = 2 \Rightarrow 1.96 < \iint_{|x|+|y| \leq 10} \frac{dx dy}{100 + \cos^2 x + \cos^2 y} < 2$$

練習題 16.5

1. (1) $D = \{(x, y) \mid 0 \leq x \leq \pi, x \leq y \leq \pi\}$

$$\begin{aligned} \iint_D \cos(x+y) dx dy &= \int_0^\pi dx \int_x^\pi \cos(x+y) dy = \int_0^\pi \sin(x+y) \Big|_{y=x}^{y=\pi} dx = \int_0^\pi -\sin x dx + \int_0^\pi \sin 2x dx \\ &= \cos x \Big|_0^\pi - \frac{1}{2} \cos 2x \Big|_0^\pi = -2 \end{aligned}$$

(2) 記 $B_1 = \{(x, y) \mid -1 \leq x \leq 1, 0 \leq y \leq 1-|x|\}$

$B_2 = \{(x, y) \mid -1 \leq x \leq 1, |x|-1 \leq y \leq 0\}$

顯然 $D = B_1 \cup B_2$, $B_1 \cap B_2 = \{(\pm 1, 0)\}$ 是零面積集

$$\begin{aligned} \Rightarrow \iint_D e^{x+y} dx dy &= \iint_{B_1} e^{x+y} dx dy + \iint_{B_2} e^{x+y} dx dy \\ &= \int_{-1}^1 dx \int_0^{1-|x|} e^{x+y} dy + \int_{-1}^1 dx \int_{|x|-1}^0 e^{x+y} dy \\ &= \int_{-1}^1 dx \int_{|x|-1}^{1-|x|} e^y dy = \int_{-1}^1 e^x (e^{1-|x|} - e^{|x|-1}) dx \\ &= 2 \int_0^1 e^x (e^{1-x} - e^{x-1}) dx = 2 \int_0^1 (e - e^{2x-1}) dx \\ &= 2e - e^{2x-1} \Big|_0^1 = 2e - e + e^{-1} = e + e^{-1} \end{aligned}$$

(3) $\iint_D x \cos(xy) dx dy = \int_{-a}^a dy \int_{-\sqrt{a^2-y^2}}^{\sqrt{a^2-y^2}} x \cos(xy) dx = \int_{-a}^a 0 dy = 0$

2. (1) 記 $D = \{(x, y) \mid 0 \leq x \leq 1, 0 \leq y \leq x^2\} = \{(x, y) \mid 0 \leq y \leq 1, \sqrt{y} \leq x \leq 1\}$

$$\int_0^1 dx \int_0^{x^2} f(x, y) dy = \int_0^1 dy \int_{\sqrt{y}}^1 f(x, y) dx$$

(2) 記 $D = \{(x, y) \mid 0 \leq x \leq 1, x^2 \leq y \leq x\} = \{(x, y) \mid 0 \leq y \leq 1, y \leq x \leq \sqrt{y}\}$

$$\int_0^1 dx \int_{x^2}^x f(x, y) dy = \int_0^1 dy \int_y^{\sqrt{y}} f(x, y) dx$$

(3) 記 $D = \{(x, y) \mid -1 \leq x \leq 1, -\sqrt{1-x^2} \leq y \leq \sqrt{1-x^2}\} = \{(x, y) \mid x^2 + y^2 \leq 1\} = \{(x, y) \mid -1 \leq y \leq 1, -\sqrt{1-y^2} \leq x \leq \sqrt{1-y^2}\}$

$$\int_{-1}^1 dx \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} f(x, y) dy = \int_{-1}^1 dy \int_{-\sqrt{1-y^2}}^{\sqrt{1-y^2}} f(x, y) dx$$

姓名：

(號)

成績

年級：

科目：

3. 記 $D = \{(x, y) \mid 0 \leq x \leq a, 0 \leq y \leq x\} = \{(x, y) \mid 0 \leq y \leq a, y \leq x \leq a\}$.

$$\int_0^a dx \int_0^x f(x) f(y) dy = \int_0^a dy \int_y^a f(x) f(y) dx = \int_0^a dx \int_x^a f(y) f(x) dy$$

$$\begin{aligned} \text{因此 } 2 \int_0^a dx \int_0^x f(x) f(y) dy &= \int_0^a dx \int_0^a f(x) f(y) dy = \left(\int_0^a f(x) dx \right) \left(\int_0^a f(y) dy \right) \\ &= \left(\int_0^a f(x) dx \right)^2 \end{aligned}$$

$$\Rightarrow \int_0^a dx \int_0^x f(x) f(y) dy = \frac{1}{2} \left(\int_0^a f(x) dx \right)^2$$

4. 沿用上題的集合 D.

$$\int_0^a dx \int_0^x f(y) dy = \iint_D f(y) dx dy = \int_0^a dy \int_y^a f(y) dx = \int_0^a f(y) (a-y) dy = \int_0^a (a-t) f(t) dt$$

5. $\iint_{x^2+y^2 \leq r^2} f(x, y) dx dy = f(\xi_x, \xi_y) \cdot \pi r^2$ (積分中值定理).

$$\frac{1}{\pi r^2} \iint_{x^2+y^2 \leq r^2} f(x, y) dx dy = f(\xi_x, \xi_y)$$

所以 $\lim_{r \rightarrow 0} f(\xi_x, \xi_y) = f(0, 0)$, 即原極限 = $f(0, 0)$